



PROJECT REPORT

Stability and Bifurcation of a Simple Food Chain in a Chemostat with Removal Rates

Course: Math Modeling
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Contents

1	Introduction	1
2	Mathematical Model and Theory	2
3	Numerical Methods and Implementation	7
4	Simulation of the Main Regimes	10
5	Theorem Validation	16
6	Sensitivity and Robustness	19
7	Conclusion	23
	References	24

1 Introduction

1.1 The Chemostat and Food-Chain Dynamics

A *chemostat* is a laboratory apparatus for the continuous culture of micro-organisms. Fresh medium containing a growth-limiting nutrient is pumped into a well-stirred vessel at a constant rate, and culture fluid — containing nutrient, organisms, and waste — is drawn off at the same rate, so that the working volume remains fixed. The constant inflow and outflow define a *dilution rate* that removes every component of the culture at a known speed. Because the operator controls both the input nutrient concentration and the dilution rate, and because both quantities are directly measurable, the chemostat is the standard experimental and theoretical model of microbial growth: it turns an otherwise complicated ecological setting into a system whose parameters are reproducible and whose dynamics can be written down explicitly.

On top of this apparatus we place a *food chain* with trophic structure. In the chain studied here a prey organism consumes the externally supplied nutrient, a first predator feeds on the prey, and a top predator feeds on the first predator. The defining feature of a *chain* (as opposed to a food web) is that each species eats only the level immediately below it: the first predator does not consume nutrient directly, and the top predator consumes neither nutrient nor prey. Energy therefore flows in one direction, from the nutrient supply up through three living levels, while the dilution flow and each organism's own mortality continually remove material from every level. The central question is how the competition between bottom-up nutrient supply and top-down removal determines which species survive in the long run.

1.2 Biological Problem and Research Questions

Earlier chemostat food-chain studies, in particular Li and Kuang [2], analysed a single prey and a single predator. El-Sheikh and Mahrouf [1] extend that work in two directions: they add a second predator (one prey, two predators), and they make the model more realistic by assigning each organism its own *removal rate*. In a real culture an organism is removed both by chemostat washout and by its own death, so its effective removal rate is the sum of the two. When these death rates differ from species to species — the biologically generic case — the removal rates are no longer equal.

This single change has a large mathematical consequence. When all organisms are removed at the same rate, the total amount of material in the vessel obeys a conservation law, and the four-variable system can be reduced to three variables. With *distinct* removal rates the conservation law fails, the reduction is no longer available, and the full four-dimensional system must be analysed directly. Against this background the reference poses the questions that this report addresses:

1. **Survival.** Under what conditions on the nutrient supply and the removal rates does each trophic level survive rather than wash out of the vessel?
2. **Coexistence and persistence.** When do all three organisms coexist, and is that coexistence *uniformly persistent* — robustly bounded away from extinction — rather than merely transient?
3. **Qualitative transitions.** As a control parameter is varied, when does a stable coexistence state lose stability and give way to sustained oscillations through a *Hopf bifurcation*?

1.3 Modelling Approach

We model the food chain as a system of continuous-time ordinary differential equations (ODEs) in which the four state variables are the concentrations of nutrient, prey, first predator, and top predator. A continuous-time deterministic description is the natural choice here for several converging reasons. First, the chemostat is genuinely a continuous-flow system, and the modelled

quantities are concentrations of large microbial populations, for which a deterministic continuum description is accurate and an agent-by-agent account is unnecessary. Second, the questions are dynamic and qualitative — who survives, who coexists, when does coexistence become periodic — and these are precisely the questions an ODE model answers through equilibria, eigenvalue stability, and bifurcation theory. Third, the model is parsimonious: it represents the whole chain with four states and a handful of interpretable parameters, yet it is rich enough to exhibit the full hierarchy of outcomes and a Hopf bifurcation. Finally, it matches the modelling tradition the paper builds on, so our results can be compared directly with the reference.

The ODE formulation encodes a specific and standard set of biological assumptions, which we make explicit because they justify the model’s form: the vessel is perfectly stirred, so every concentration is spatially uniform; each organism consumes only the level directly beneath it; the input nutrient concentration and the dilution rate are held fixed; each organism carries its own removal rate; and the per-capita growth responses are smooth, start from zero when food is absent, and increase with the available food. These assumptions are formalised in Section 2.

1.4 Scope and Contribution

The scope is deliberately bounded. We study the four-dimensional chemostat food chain with Monod (Michaelis–Menten) growth responses and three distinct removal rates, restricted to nonnegative concentrations. Spatial structure, time delays, stochastic fluctuations, and more than two predators are outside our scope.

The reference is a purely theoretical paper: it proves theorems but reports no experiments. Our contribution is therefore to *reproduce and demonstrate its results numerically*. We first restate, in one consistent notation, the model and the core theorems we will study (Section 2). We then describe the numerical methods used (Section 3), simulate the main biological regimes and compare them with the analytical equilibria (Section 4), validate the central theorems — local stability, uniform persistence, and the Hopf bifurcation (Section 5) — and finally quantify the sensitivity and numerical robustness of the conclusions (Section 6), before concluding (Section 7).

2 Mathematical Model and Theory

This section sets up the model in one consistent notation and states the theoretical results of El-Sheikh and Mahrouf [1] that the remainder of the report reproduces. We keep the development self-contained but emphasise the results at the heart of the paper — the equilibria and their stability, uniform persistence of coexistence, and the Hopf bifurcation.

2.1 Nondimensional Model

The reference first writes the chain in dimensional variables and then rescales: nutrient is measured relative to the input concentration, time in units of the inverse dilution rate, and the organism concentrations using the corresponding yield constants. This nondimensionalisation fixes the nutrient inflow and washout terms at one and reduces the model to four state variables and three effective removal rates. The state vector is $X(t) = (S(t), x(t), y(t), z(t))^T \in \mathbb{R}_+^4$ with the meanings of Table 1, and the model is

$$\begin{aligned} \frac{dS}{dt} &= 1 - S - f_1(S) x, \\ \frac{dx}{dt} &= x(f_1(S) - D_1) - f_2(x) y, \\ \frac{dy}{dt} &= y(f_2(x) - D_2) - f_3(y) z, \\ \frac{dz}{dt} &= z(f_3(y) - D_3), \end{aligned} \quad S(0), x(0), y(0), z(0) > 0. \tag{1}$$

Table 1: State variables of the nondimensional chemostat food-chain model.

Symbol	Biological meaning	Domain
$S(t)$	Nutrient concentration	$S \geq 0$
$x(t)$	Prey concentration; consumes the nutrient	$x \geq 0$
$y(t)$	First-predator concentration; consumes x	$y \geq 0$
$z(t)$	Top-predator concentration; consumes y	$z \geq 0$

Each equation is a balance. The first balances normalised nutrient inflow, nutrient washout, and consumption by the prey. The prey grows at per-capita rate $f_1(S)$, is removed at rate D_1 , and is consumed by the first predator through the flux $f_2(x)y$; the chain continues upward. The structure — y does not consume nutrient and z consumes neither nutrient nor prey — means the transfer terms do not cancel under summation once the removal rates differ, which is exactly why the conservation-law reduction is lost and the full four-dimensional system must be studied.

Each $D_i > 0$ is the dimensionless total removal rate of one organism (chemostat washout plus a species-specific death rate, divided by the dilution rate). The nine numerical parameters are therefore $(a_1, k_1, a_2, k_2, a_3, k_3, D_1, D_2, D_3)$.

2.2 Response Functions and Assumptions

The reference allows general response functions f_i subject to the assumptions that each $f_i : \mathbb{R}_+ \rightarrow \mathbb{R}_+$ is continuously differentiable, $f_i(0) = 0$, and $f'_i(u) > 0$ for $u > 0$; that is, growth vanishes when food is absent and increases with the available food. For concrete, reproducible simulations we specialise to the Monod (Michaelis–Menten) form

$$f_i(u) = \frac{a_i u}{k_i + u}, \quad f'_i(u) = \frac{a_i k_i}{(k_i + u)^2}, \quad f_i^{-1}(d) = \frac{d k_i}{a_i - d} \quad (0 \leq d < a_i), \quad (2)$$

where $a_i > 0$ is the maximal specific growth rate and $k_i > 0$ the half-saturation constant, since $f_i(k_i) = a_i/2$. Monod kinetics is the canonical microbial growth law: growth rises almost linearly when food is scarce and saturates when food is plentiful, with only two interpretable parameters per trophic level. The derivative and inverse in (2) are exactly what the equilibrium and stability calculations require. A finite positive equilibrium needs $D_i < a_i$ so that $f_i^{-1}(D_i)$ exists; stronger feasibility conditions are derived below.

Remark 1 *The printed nondimensional system in [1] contains typographical inconsistencies with its own Jacobian and equilibrium formulas. We use the analysis-consistent form (1), which reproduces those formulas, and we likewise follow the paper’s washout-stability theorem rather than the opposite sign that appears in its closing discussion.*

2.3 Non-negativity, Boundedness, and Dissipativity

The model is well posed as a population model. The nonnegative orthant is positively invariant: at $S = 0$ the first equation gives $\frac{dS}{dt} = 1 > 0$, and at $x = 0$, $y = 0$, or $z = 0$ the corresponding derivative vanishes because $f_i(0) = 0$, so a solution starting with nonnegative concentrations cannot become negative. Summing (1) gives, for the total mass $T = S + x + y + z$,

$$\frac{dT}{dt} = 1 - S - D_1 x - D_2 y - D_3 z, \quad \text{hence} \quad 1 - D_{\max} T \leq \frac{dT}{dt} \leq 1 - D_{\min} T,$$

with $D_{\max} = \max\{1, D_1, D_2, D_3\}$ and $D_{\min} = \min\{1, D_1, D_2, D_3\}$. The scalar comparison principle then bounds T between $1/D_{\max}$ and $1/D_{\min}$ in the long run, which is the content of the following result.

Theorem 1 (Boundedness and dissipativity; [1, Thm. 2.1]) Let $D_{\max} = \max\{1, D_1, D_2, D_3\}$, $D_{\min} = \min\{1, D_1, D_2, D_3\}$, and for a constant $k > 0$ define the region

$$\Gamma = \left\{ X \in \mathbb{R}_+^4 : \frac{1}{D_{\max}} - k \leq S + x + y + z \leq \frac{1}{D_{\min}} + k \right\}.$$

Then (i) Γ is positively invariant; (ii) every solution of (1) with nonnegative initial data is eventually uniformly bounded and attracted into Γ , so that the total mass $T = S + x + y + z$ obeys

$$\frac{1}{D_{\max}} \leq \liminf_{t \rightarrow \infty} T(t) \leq \limsup_{t \rightarrow \infty} T(t) \leq \frac{1}{D_{\min}};$$

and (iii) the system (1) is dissipative in the sense that a single bounded set eventually absorbs every trajectory.

Dissipativity is both biologically necessary — no population can grow without bound in a finite vessel — and numerically useful: an unbounded computed trajectory signals an invalid parameter or an integration failure rather than genuine model behaviour.

2.4 Equilibria and Their Existence

An equilibrium $P = (S^*, x^*, y^*, z^*)$ satisfies the right-hand side of (1) equal to zero. Deciding which organism concentrations vanish yields four trophic regimes.

Washout P_0 . If all organisms are absent, nutrient converges to its normalised input level: $P_0 = (1, 0, 0, 0)$, which always exists.

Prey-only P_1 . For $x_1 > 0$, $y_1 = z_1 = 0$, the prey equation requires $f_1(S_1) = D_1$, so

$$P_1 = (S_1, x_1, 0, 0), \quad S_1 = f_1^{-1}(D_1) = \frac{D_1 k_1}{a_1 - D_1}, \quad x_1 = \frac{1 - S_1}{D_1}.$$

Feasibility $0 < S_1 < 1$ is equivalent to $D_1 < f_1(1) = a_1/(1 + k_1)$ — precisely the condition that prey can increase when rare at P_0 .

Two-species P_2 . For prey and the first predator with $z = 0$, the y -equation requires $f_2(x_2) = D_2$, so

$$P_2 = (S_2, x_2, y_2, 0), \quad x_2 = f_2^{-1}(D_2), \quad S_2 + f_1(S_2)x_2 = 1, \quad y_2 = \frac{x_2(f_1(S_2) - D_1)}{D_2}.$$

Since $S \mapsto S + f_1(S)x_2$ is strictly increasing from 0 to a value greater than one, S_2 is the unique root in $(0, 1)$; feasibility $y_2 > 0$ requires $f_1(S_2) > D_1$.

Coexistence P_3 . At an interior equilibrium all components are positive. The top-predator equation fixes $y_3 = f_3^{-1}(D_3) = D_3 k_3/(a_3 - D_3)$, and the remaining coordinates satisfy

$$S_3 + f_1(S_3)x_3 = 1, \quad x_3(f_1(S_3) - D_1) = f_2(x_3)y_3, \quad z_3 = \frac{y_3(f_2(x_3) - D_2)}{D_3}.$$

These coupled equations have no useful closed form even for Monod responses; the state is feasible only when all coordinates are positive, which in particular needs $f_2(x_3) > D_2$.

The invasion hierarchy. The equilibria form a trophic ladder governed by the *invasion rates* — the per-capita growth of a missing species introduced at low concentration,

$$\lambda_x = f_1(1) - D_1, \quad \lambda_y = f_2(x_1) - D_2, \quad \lambda_z = f_3(y_2) - D_3. \quad (3)$$

A positive rate means the next level can invade the lower equilibrium; a negative rate means it washes out. Because each f_i is strictly increasing, every defining equation has a unique root, so each equilibrium, when feasible, is unique. Existence of the boundary equilibria is established in the reference by a persistence argument: if P_1 is a hyperbolic rest point repelling in the y -direction and the boundary flow has no cycles, the (S, x, y) subsystem is uniformly persistent, so P_2 exists [1, Lem. 2.3, Thm. 2.6–2.7]. Table 2 summarises the conditions, and Table 3 gives representative computed equilibria.

Table 2: Equilibrium hierarchy and feasibility conditions.

Equilibrium	Surviving organisms	Main feasibility / invasion condition
P_0	none	always exists
P_1	x	$\lambda_x > 0$
P_2	x, y	P_1 exists and $\lambda_y > 0$
P_3	x, y, z	P_2 exists, $\lambda_z > 0$, and P_3 is positive

Table 3: Representative equilibria from the four governing regimes (computed for the parameter sets of Section 4).

Regime	Richest equilibrium	Coordinates (S, x, y, z)
Washout	P_0	$(1, 0, 0, 0)$
Prey only	P_1	$(0.166667, 1.666667, 0, 0)$
Two species	P_2	$(0.628667, 0.333333, 0.409333, 0)$
Coexistence	P_3	$(0.055067, 0.804597, 0.280534, 0.081550)$

2.5 Local Stability and Threshold Conditions

Linearising (1) at a state (S, x, y, z) gives the Jacobian

$$J = \begin{pmatrix} -1 - f'_1(S)x & -f_1(S) & 0 & 0 \\ f'_1(S)x & f_1(S) - D_1 - f'_2(x)y & -f_2(x) & 0 \\ 0 & f'_2(x)y & f_2(x) - D_2 - f'_3(y)z & -f_3(y) \\ 0 & 0 & f'_3(y)z & f_3(y) - D_3 \end{pmatrix}. \quad (4)$$

An equilibrium is locally asymptotically stable when every eigenvalue of J has negative real part. The Jacobian is block-triangular in the trophic sense, so at a boundary equilibrium its spectrum is the union of the eigenvalues of the surviving internal block and the invasion rates (3) of the missing species. This structure makes the stability thresholds transparent.

Stability of P_0 . At washout the eigenvalues are -1 , $f_1(1) - D_1$, $-D_2$, $-D_3$, so P_0 is locally asymptotically stable iff $\lambda_x = f_1(1) - D_1 < 0$, i.e. $D_1 > f_1(1) = a_1/(1 + k_1)$. If the inequality is reversed, prey can invade and P_0 is unstable in the x -direction.

Stability of P_1 . The (S, x) block of $J(P_1)$ has negative trace and positive determinant, so both of its eigenvalues have negative real part; the remaining two eigenvalues are $\lambda_y = f_2(x_1) - D_2$ and $-D_3$. Hence P_1 is locally asymptotically stable iff $\lambda_y < 0$; when $\lambda_y > 0$ it is a saddle repelling trajectories in the y -direction.

Stability of P_2 . The characteristic polynomial of $J(P_2)$ factors as $(\lambda - \lambda_z)(\lambda^3 + a_1\lambda^2 + a_2\lambda + a_3)$, where the cubic is the characteristic polynomial of the active (S, x, y) block and $\lambda_z = f_3(y_2) - D_3$ is the transverse top-predator eigenvalue. By the Routh–Hurwitz criterion the cubic is stable iff

$$a_1 > 0, \quad a_3 > 0, \quad a_1a_2 > a_3. \quad (5)$$

Therefore P_2 is locally asymptotically stable iff the block is Routh–Hurwitz stable and $\lambda_z < 0$; if the block is stable but $\lambda_z > 0$, then P_2 is a saddle repelling in the z -direction. These saddle structures are the engine of the trophic ladder: each lower equilibrium is unstable precisely in the direction of the next predator, which is what allows the chain to build up to coexistence.

Stability of P_3 . At coexistence the characteristic polynomial is the quartic $\lambda^4 + d_1\lambda^3 + d_2\lambda^2 + d_3\lambda + d_4$, and by the Routh–Hurwitz criterion P_3 is locally asymptotically stable iff

$$d_1, d_3, d_4 > 0 \quad \text{and} \quad \Phi := d_1d_2d_3 - d_3^2 - d_1^2d_4 > 0. \quad (6)$$

The quantity Φ plays the decisive role in the bifurcation analysis below.

These results give a direct, computable stability test at every equilibrium and are collected as the local-stability theorem of the reference.

Theorem 2 (Local stability of the equilibria; [1, Thm. 2.2, 2.4, 2.8]) *Let $\lambda_x, \lambda_y, \lambda_z$ be the invasion rates (3).*

- (i) *The washout equilibrium P_0 is locally asymptotically stable if and only if $\lambda_x < 0$ (equivalently $f_1(1) < D_1$); its Jacobian eigenvalues are $-1, \lambda_x, -D_2, -D_3$.*
- (ii) *The prey-only equilibrium P_1 is locally asymptotically stable if and only if $\lambda_y < 0$; the two eigenvalues of its (S, x) block always have negative real part, while the remaining eigenvalues are λ_y and $-D_3$.*
- (iii) *The two-species equilibrium P_2 is locally asymptotically stable if and only if the cubic factor of its characteristic polynomial satisfies the Routh–Hurwitz conditions (5) and the transverse eigenvalue satisfies $\lambda_z < 0$.*

Theorem 3 (Boundary saddles; [1, Thm. 2.5, 2.9]) *If $\lambda_y > 0$, then P_1 is a hyperbolic saddle point that repels trajectories in both the y - and z -directions, with two-dimensional stable and unstable manifolds. If the (S, x, y) block of P_2 is Routh–Hurwitz stable but $\lambda_z > 0$, then P_2 is a hyperbolic saddle point that repels in the z -direction, with a one-dimensional unstable manifold. These transverse instabilities are the mechanism that pushes the system up the trophic ladder: each lower equilibrium is unstable precisely in the direction of the next predator.*

2.6 Uniform Persistence and Coexistence

Local stability describes trajectories close to one equilibrium; uniform persistence concerns all positive solutions at once. The reference combines the dissipativity of Theorem 1 with boundary-flow arguments: the boundary equilibria P_1 and P_2 are shown to be globally asymptotically stable on their respective subsystems (established with Liapunov functions), and, when they additionally repel the missing species, no strictly positive trajectory can approach the boundary.

Theorem 4 (Uniform persistence; [1, Thm. 3.4]) *Assume that*

- (i) *P_1 is a hyperbolic saddle repelling in the y - and z -directions, and P_2 is a hyperbolic saddle repelling in the z -direction (Theorem 3);*
- (ii) *the system is dissipative and its solutions are eventually uniformly bounded (Theorem 1); and*

- (iii) P_1 and P_2 are globally asymptotically stable on the boundary subsystems $\mathbb{R}_{Sx}^+$ and $\mathbb{R}_{Sxy}^+$ respectively — a property the reference establishes by constructing Liapunov functions.

Then system (1) is uniformly persistent: there exists $\eta > 0$ such that

$$\liminf_{t \rightarrow \infty} S(t), \liminf_{t \rightarrow \infty} x(t), \liminf_{t \rightarrow \infty} y(t), \liminf_{t \rightarrow \infty} z(t) \geq \eta$$

for every solution with positive initial data. In particular, the boundary flow cannot attract any interior trajectory, and a positive interior equilibrium P_3 exists.

In words, when each lower equilibrium lets the next predator invade, every population is robustly bounded away from extinction regardless of the starting state, and a genuine coexistence state must exist.

2.7 Hopf Bifurcation

The paper's title result concerns how a *stable* coexistence state can lose stability and begin to oscillate. The reference introduces a real bifurcation parameter μ into the prey response, $f_1 = f_1(S; \mu)$, and studies how the spectrum of $J(P_3)$ moves as μ varies. A Hopf bifurcation requires a complex-conjugate pair of eigenvalues to cross the imaginary axis with nonzero speed (the transversality condition) while the other eigenvalues stay away from it. In terms of the quartic coefficients this means reaching the Routh–Hurwitz boundary

$$\Phi = d_1 d_2 d_3 - d_3^2 - d_1^2 d_4 = 0 \quad (7)$$

with $d_1, d_3, d_4 > 0$: as Φ passes through zero, the conjugate pair moves from the left half-plane ($\Phi > 0$, stable) to the right ($\Phi < 0$, unstable).

Theorem 5 (Hopf bifurcation at P_3 ; [1, Thm. 4.3]) *Consider system (1) with $f_1 = f_1(S; \mu)$ depending on the bifurcation parameter μ , and assume that*

- (i) *the system is uniformly persistent (Theorem 4);*
- (ii) *the interior equilibrium $P_3(\mu)$ exists; and*
- (iii) *along the family the quartic coefficients remain positive, $d_1, d_3, d_4 > 0$, while the Routh–Hurwitz quantity Φ of (7) changes sign transversally at some value $\mu = \mu^*$, i.e. $\Phi(\mu^*) = 0$ and $\Phi'(\mu^*) \neq 0$.*

Then at $\mu = \mu^$ a complex-conjugate pair of eigenvalues of $J(P_3)$ crosses the imaginary axis with nonzero speed, and (1) undergoes a Hopf–Andronov–Poincaré bifurcation at P_3 , giving rise to a family of periodic (oscillatory) coexistence solutions in the positive orthant. The reference establishes an analogous bifurcation at P_2 under the additional hypotheses $f_2'(x_2)x_2 > D_2$ and $f_1(S_2; \mu) > D_1$, and shows that no Hopf bifurcation can occur at the saddle P_1 [1, Cor. 4.1, Thm. 4.2].*

The core results — dissipativity (Theorem 1), the local-stability law (Theorem 2), uniform persistence (Theorem 4), and the Hopf bifurcation (Theorem 5) — are exactly what Sections 4 and 5 reproduce numerically.

3 Numerical Methods and Implementation

The analytical results of Section 2 are reproduced by a small, transparent set of numerical methods. This section describes those methods — the integration scheme, the simulation algorithm and its safeguards, the equilibrium and eigenvalue computations, the bifurcation diagnostics, and the sensitivity measure. All routines are deterministic and were implemented from first principles using only standard numerical building blocks, so that every figure in the report is exactly reproducible. Throughout, the parameter sets are fixed representative values chosen to realise the regimes of interest; they are neither fitted to data nor generated randomly.

3.1 The Numerical Initial-Value Problem

For a parameter vector $\theta = (a_1, k_1, a_2, k_2, a_3, k_3, D_1, D_2, D_3)$ the simulator solves the initial-value problem

$$\frac{dX}{dt} = F(X; \theta), \quad X(0) = X_0 = (S_0, x_0, y_0, z_0)^T, \quad (8)$$

where F is the right-hand side of (1) with Monod responses (2). The numerical problem is thus not a different model from the theory but a time-discretised approximation of the same four-dimensional system. The integrator expects a finite positive step Δt and a finite nonnegative horizon T ; unless stated otherwise the main simulations use the standard start $X_0 = (0.5, 0.2, 0.1, 0.05)$.

3.2 The Vector Field

The core routine evaluates the four derivatives

$$F_1 = 1 - S - f_1(S)x, \quad F_2 = x(f_1(S) - D_1) - f_2(x)y, \quad F_3 = y(f_2(x) - D_2) - f_3(y)z, \quad F_4 = z(f_3(y) - D_3).$$

The three response functions are stored together with their analytic derivatives and inverses (2), which the equilibrium and stability computations reuse. Inside a response evaluation a negative food argument is treated as zero; this is a safeguard for the intermediate trial states of an explicit integrator and does not alter the continuous model, whose nonnegative orthant is invariant.

3.3 Time Integration

Two explicit fixed-step schemes are used. If X_n approximates $X(t_n)$ and $h = \Delta t$, the first-order *explicit Euler* step is

$$X_{n+1} = X_n + h F(X_n; \theta), \quad (9)$$

and the classical fourth-order *Runge-Kutta* (RK4) step is

$$X_{n+1} = X_n + \frac{h}{6}(K_1 + 2K_2 + 2K_3 + K_4), \quad (10)$$

with $K_1 = F(X_n)$, $K_2 = F(X_n + \frac{h}{2}K_1)$, $K_3 = F(X_n + \frac{h}{2}K_2)$, $K_4 = F(X_n + hK_3)$. Euler is first-order, so halving the step roughly halves the global error; RK4 is fourth-order for smooth solutions and is therefore far more accurate at the same step. RK4 is used for all reported biological simulations; Euler is retained as a diagnostic baseline in the robustness study of Section 6: if a coarse Euler step gives a different qualitative regime from RK4, that difference is numerical error rather than a new biological conclusion.

3.4 The Simulation Algorithm

The solver loop is deliberately simple and reproducible. For a chosen scenario it

1. reads the nine parameters and builds the response functions and removal rates;
2. initialises the time list at $t_0 = 0$ and the state list at X_0 ;
3. repeats the Euler or RK4 update until the horizon T is reached;
4. clamps any completed component that has become negative back to zero and counts the clamp;
5. stops early if any component exceeds an explosion threshold of 10^{10} ;
6. records the trajectory together with a diagnostic summary.

The nonnegative orthant is invariant for the exact ODE, but in floating-point arithmetic an explicit method can produce a tiny negative value near a boundary. Clamping prevents such round-off from being misread as a negative concentration; because every clamp is counted, the diagnostics still reveal whether a run relied on the safeguard. This turns the abstract question “is the integrator stable for this step?” into an automatic, quantitative flag.

3.5 Diagnostics and Regime Classification

For each completed run the diagnostics record the final state, the componentwise minima and maxima, the number of clamps, the explosion flag, the number of steps, and the Euclidean distances to every existing equilibrium. A run is deemed to have *converged* when the componentwise standard deviation over the last 10% of the trajectory falls below 10^{-6} .

The long-run regime is read from the final state with a small threshold $\varepsilon = 10^{-3}$: the regime is P_0 if $x \leq \varepsilon$, else P_1 if $y \leq \varepsilon$, else P_2 if $z \leq \varepsilon$, else P_3 . The nearest analytical equilibrium is also recorded by Euclidean distance, which is a more reliable classifier than visual inspection when a component is small or when damped oscillations remain visible. For an oscillatory run a single final point is insufficient, so the trajectory is additionally summarised by late-window means, counts of local maxima in the predator variables, and the ratio of late to mid-window amplitudes; an approach is classified as a damped oscillation toward P_3 only when its late mean is close to P_3 , it shows repeated predator peaks after the transient, and its late amplitude is smaller than its mid-window amplitude.

3.6 Equilibrium Computation

The equilibria of Section 2.4 are obtained algebraically rather than by long-time integration, so that they serve as independent targets for the simulations. The Monod inverse (2) gives the direct coordinates S_1, x_2, y_3 in closed form. The coupled scalar equations — $S + f_1(S)x = 1$ for the nutrient balance and the prey-balance residual that fixes x_3 — are solved by bisection, which is well posed because each defining map is strictly monotone, guaranteeing a unique bracketed root. Feasibility (positivity of every coordinate) is checked before an equilibrium is accepted.

3.7 Stability and Bifurcation Diagnostics

Local stability is decided from the eigenvalues of the Jacobian (4) at each equilibrium. The characteristic polynomial is formed from the Jacobian by the Faddeev–Le Verrier recursion, and its roots — the eigenvalues — are then extracted by a standard simultaneous root-finding iteration. The sign of the largest real part gives the stability classification (stable node or focus, saddle, unstable node or focus), which is cross-checked against the analytic invasion thresholds (3) and the Routh–Hurwitz conditions (5)–(6).

For the Hopf analysis a bifurcation parameter μ is swept while the largest real part of the spectrum and the Routh–Hurwitz quantity Φ of (7) are tracked. A simultaneous sign change of both, with a nonzero slope of $\max \operatorname{Re} \lambda$ in μ , identifies a transversal eigenvalue crossing. The nonlinear consequence is confirmed by integrating to a long horizon and measuring the amplitude of the late-time oscillation, which is essentially zero below the crossing and grows continuously above it.

For uniform persistence (Theorem 4) the persistence margin is estimated as the late-time minimum of each component, taken over a spread of initial conditions; a strictly positive margin certifies that every level stays bounded away from extinction.

3.8 Sensitivity by Finite Differences

The sensitivity of an equilibrium coordinate q to a parameter p is measured by the dimensionless *elasticity*

$$\mathcal{S}(q, p) = \frac{p}{q} \frac{\partial q}{\partial p} \approx \frac{q(p(1 + \delta)) - q(p(1 - \delta))}{2 q \delta}, \quad (11)$$

a central finite difference with a small relative perturbation δ (we use $\delta = 1\%$) that is second-order accurate. The dimensionless form makes parameters with different units directly comparable: $\mathcal{S}(q, p) = 2.5$ means a 1% increase in p raises q by 2.5%. Because the elasticities are taken from

the exact equilibrium solver rather than from a long simulation, the perturbation response is clean and free of integration noise.

3.9 Reproducibility

Every experiment reads the same fixed parameter table and records its integrator, step size, horizon, and initial state alongside the numerical output. The methods above were also checked by a small suite of unit tests covering the vector field (a known stationary value, no mutation of the input, the correct sign on the coordinate boundaries), the solvers (the Euler recurrence, RK4’s fourth-order convergence against a scalar benchmark, Euler’s first order), the equilibrium formulas, the eigenvalues at washout, and the diagnostics. These tests do not prove the biological model, but they remove the main implementation risks — sign errors in the vector field, incorrect time stepping, silent negative concentrations, and disagreement between the simulator and the equilibrium formulas — so that the results that follow can be read as properties of the model rather than artefacts of the computation.

4 Simulation of the Main Regimes

The purpose of this section is to connect the analytical equilibria of Section 2 with the visible long-time dynamics of the model, and to confirm that the simulator reproduces the equilibrium structure to high accuracy. We integrate five representative parameter sets, one for each biological regime, and compare each final state with its analytical target.

4.1 Experimental Setup

All trajectories are computed with the fourth-order Runge–Kutta scheme (10) at step $\Delta t = 0.01$ from the common initial condition $X_0 = (0.5, 0.2, 0.1, 0.05)$, so that any difference in outcome is attributable to the parameters rather than to the starting state. The four equilibrium scenarios are integrated to $T = 500$; the oscillatory scenario is integrated to $T = 2000$, because its transient oscillations decay slowly and a shorter run could give a misleading classification. No run exploded and, for these scenarios, no run required non-negativity clamping. The five parameter sets are listed in Table 4, and an overview of the resulting regimes appears in Figure 1.

Table 4: Parameter sets used in the main simulation experiments.

Scenario	a_1	k_1	a_2	k_2	a_3	k_3	D_1	D_2	D_3
P_0 washout	0.60	0.50	1.50	0.50	1.20	0.40	0.80	0.70	0.70
P_1 prey only	2.00	0.50	0.50	1.00	1.00	1.00	0.50	0.60	0.60
P_2 two species	2.00	0.50	2.00	1.00	0.50	1.00	0.50	0.50	0.70
P_3 coexistence	4.80	0.17	4.30	1.85	4.60	1.66	0.72	1.11	0.665
Oscillatory response	6.52	1.76	5.07	2.64	3.78	0.0466	1.14	0.778	0.488

Main chemostat simulation scenarios

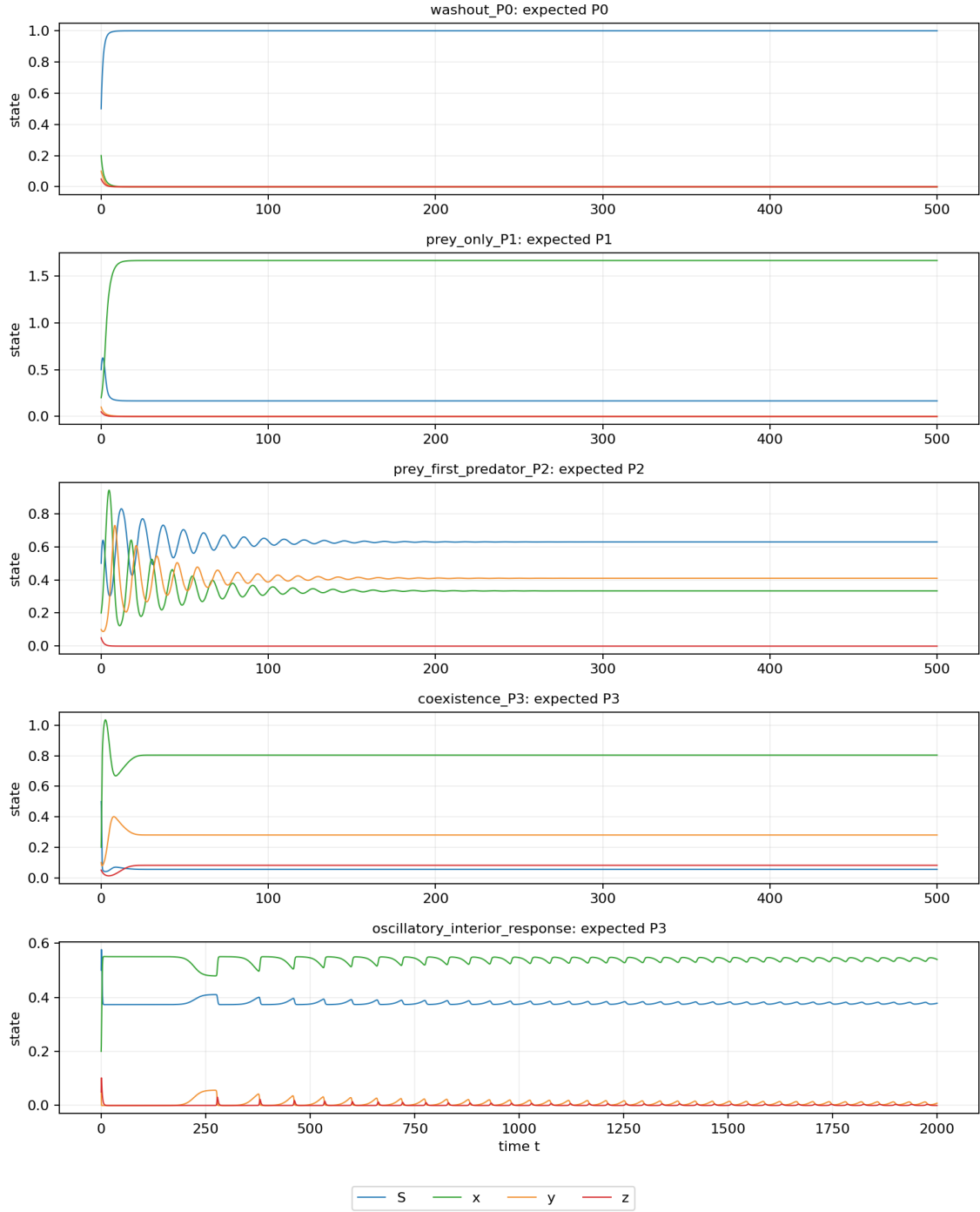


Figure 1: Overview of the five simulation regimes generated by the RK4 simulator: washout P_0 , prey-only P_1 , two-species P_2 , coexistence P_3 , and a slowly damped oscillatory approach to an interior equilibrium.

4.2 Washout Regime P_0

In the washout scenario the solution converges to $(S, x, y, z) \approx (1, 0, 0, 0)$, with distance 5.55×10^{-15} from the analytical equilibrium P_0 (Figure 2). Biologically, the prey cannot grow fast enough on the incoming nutrient to overcome its removal rate ($f_1(1) < D_1$), so it disappears; once the prey is gone, both predators lose their food and also vanish, and the nutrient returns to its normalised input concentration. This is exactly the washout-stability condition of Theorem 2.

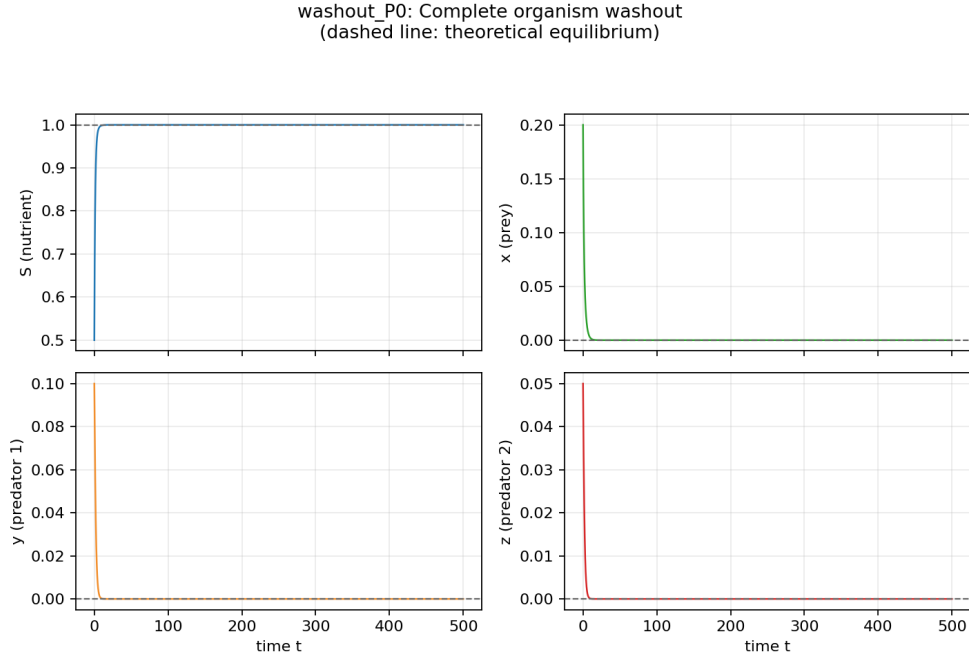


Figure 2: Time series for the washout scenario P_0 : all organisms decay and the nutrient relaxes to its input level.

4.3 Prey-Only Regime P_1

Here the final state is $(0.166667, 1.666667, 0, 0)$, at distance 2.57×10^{-14} from P_1 (Figure 3). The prey survives because it can exploit the nutrient supply, but the resulting prey concentration is not high enough for the first predator to overcome its own removal rate ($\lambda_y < 0$), so both predators wash out. The nutrient level is far lower than in the washout regime, reflecting continuous consumption by the surviving prey.

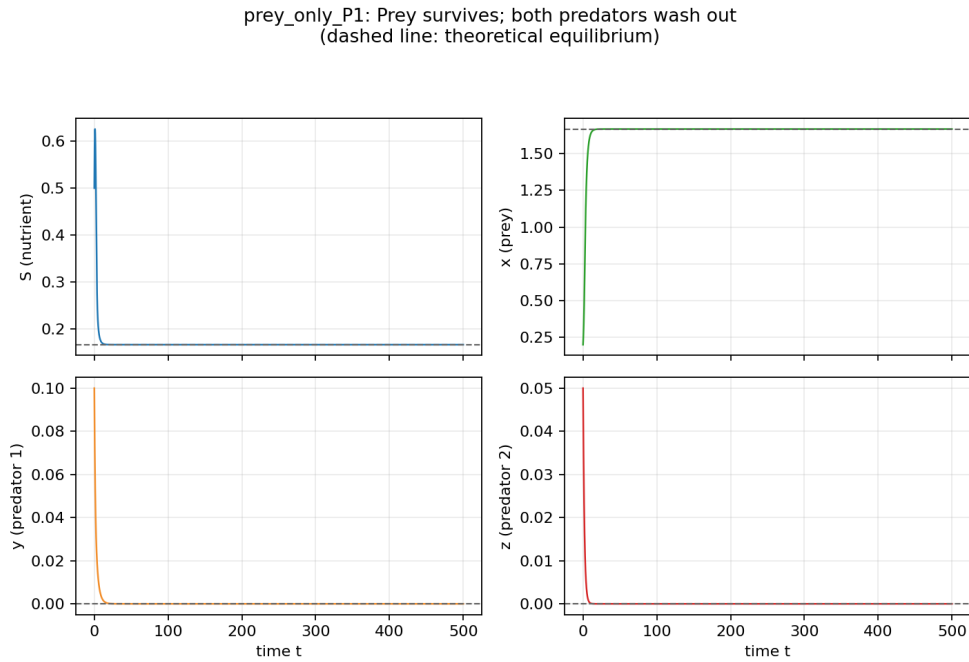


Figure 3: Time series for the prey-only scenario P_1 : prey persists while both predators decay.

4.4 Two-Species Regime P_2

The solution approaches $(0.628666, 0.333334, 0.409333, 0)$, which is 1.29×10^{-6} from P_2 (Figure 4). The prey and first predator persist while the top predator is removed: the first predator can invade the prey-only equilibrium ($\lambda_y > 0$), but the second predator cannot invade the prey–predator equilibrium ($\lambda_z < 0$). The approach contains small damped transients, a reminder that temporary oscillation should not be confused with sustained periodic behaviour.

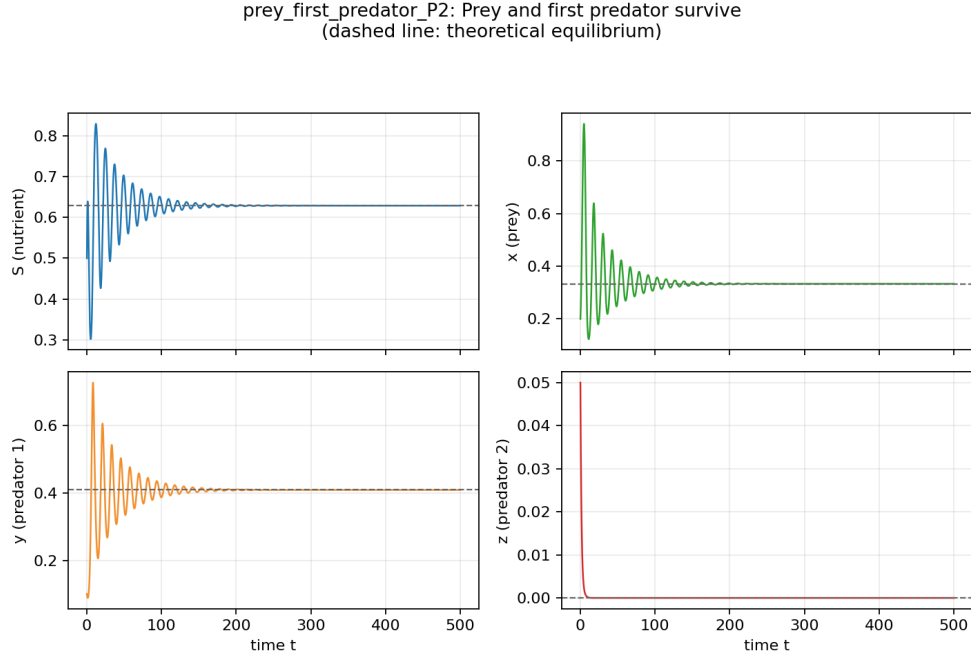


Figure 4: Time series for the two-species scenario P_2 : prey and first predator coexist while the top predator is excluded.

4.5 Coexistence Regime P_3

In the coexistence scenario all four variables converge to positive values, $(0.055067, 0.804597, 0.280534, 0.081550)$, at distance 1.94×10^{-11} from the analytical coexistence equilibrium P_3 (Figure 5). This close agreement shows that the simulator reproduces the interior equilibrium for the selected parameters. Biologically, nutrient enters continuously, the prey consumes nutrient, the first predator consumes prey, and the top predator consumes the first predator, all at balanced standing stocks. A single trajectory is, of course, not a proof of uniform persistence for all admissible starts; that statement is examined directly in Section 5.

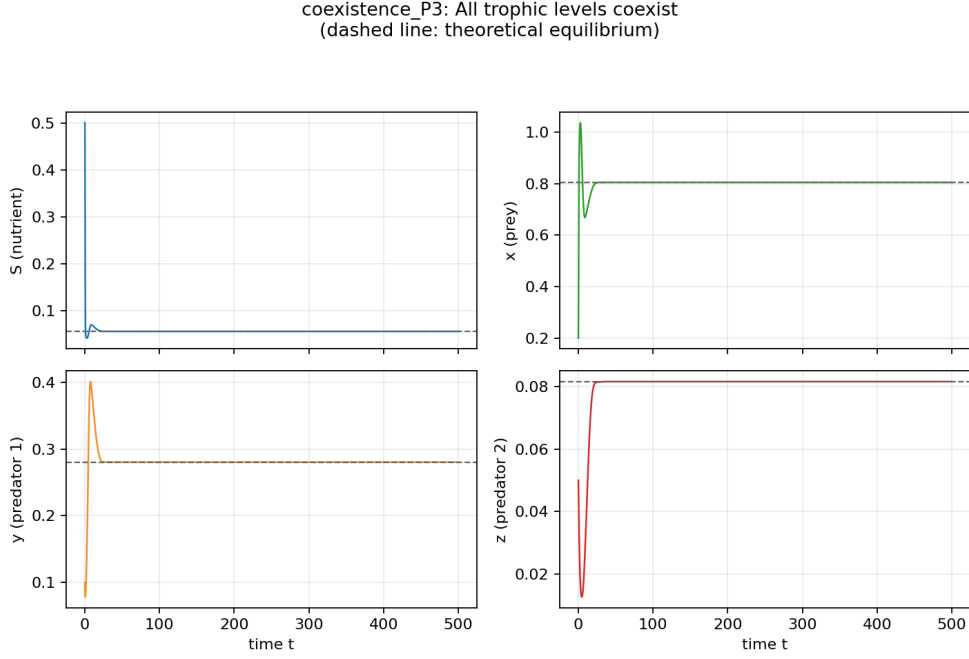


Figure 5: Time series for the coexistence scenario P_3 : all four levels settle at positive equilibrium values.

4.6 Oscillatory and Damped-Oscillatory Response

The fifth scenario was chosen to display an interior response with visible oscillations near a stability boundary. Its analytical target is $P_3 = (0.377311, 0.540994, 0.006908, 0.001193)$. At $T = 2000$ the instantaneous state is $(0.377832, 0.539609, 0.008618, 0.000146)$, at distance 2.49×10^{-3} from P_3 — larger than in the previous cases because the endpoint depends on the phase of the residual oscillation. To avoid misclassification we therefore use window statistics: over the late window $1500 \leq t \leq 2000$ the mean state is only 4.66×10^{-4} from P_3 ; after $t = 500$ the trajectory shows 23 local maxima in y and 20 in z ; and the ratio of late-window to mid-window amplitude is 0.425, so the oscillations are decaying. These measurements identify the behaviour as a slowly damped oscillatory approach to P_3 (Figures 6–7). They are consistent with dynamics near a Hopf stability boundary, but they do not by themselves prove a Hopf bifurcation; that bifurcation is established directly in Section 5.

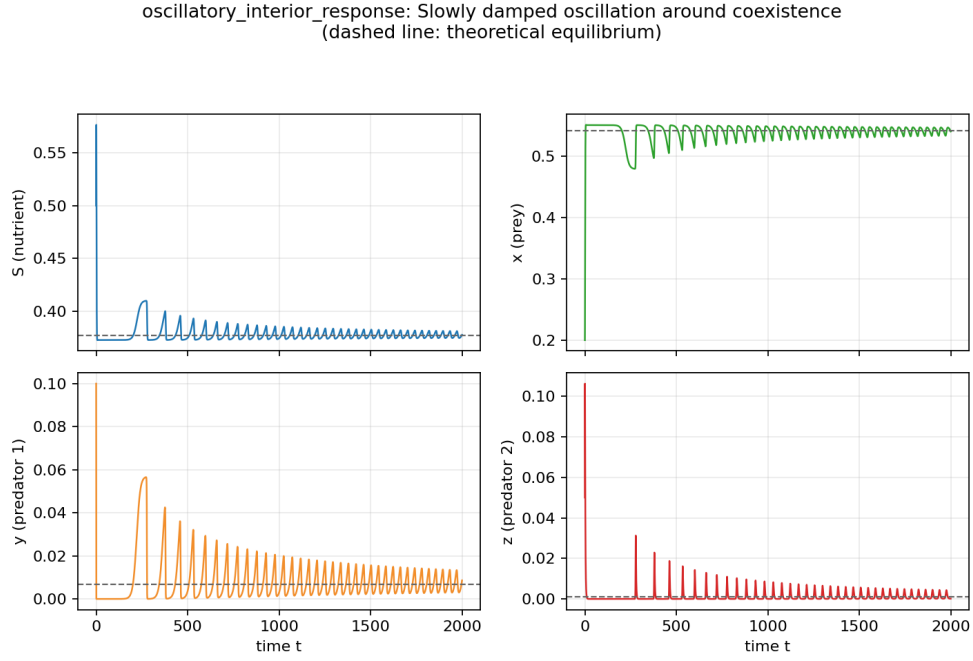


Figure 6: Time series for the slowly damped oscillatory scenario: the interior state is approached through decaying oscillations.

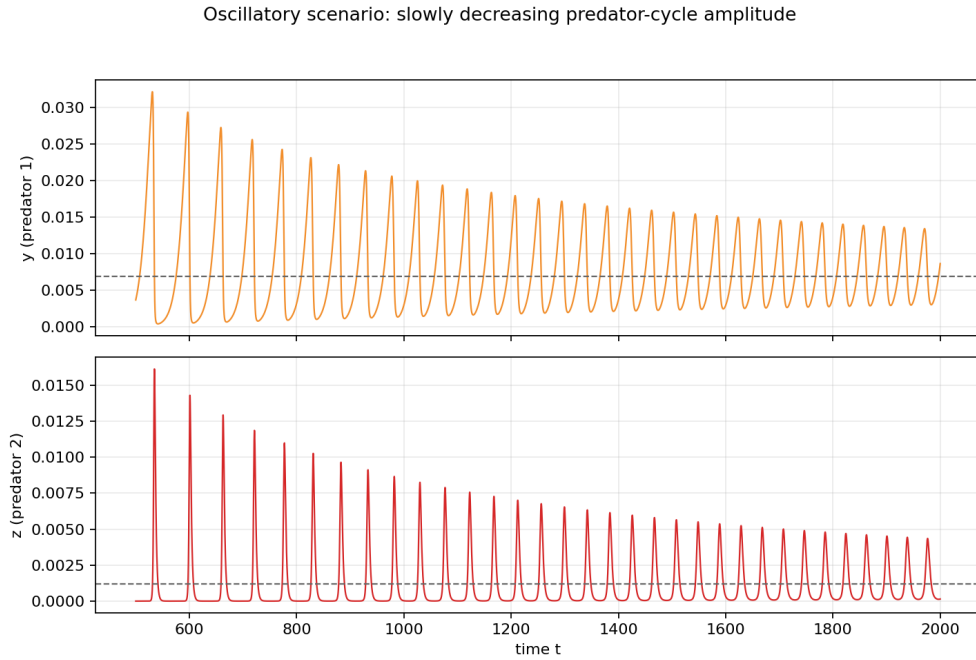


Figure 7: Zoom of the predator oscillations after the initial transient, showing the slow decay of the amplitude.

4.7 Numerical versus Theoretical Equilibria

Table 5 compares the numerical final states with their analytical targets. The four equilibrium scenarios agree with the predicted equilibria to between six and fifteen significant figures; the oscillatory case is naturally less well represented by a single endpoint, so its late-window mean is used as the measure of convergence. Taken together, the five regimes confirm the trophic

survival hierarchy — in P_0 the whole chain washes out; in P_1 the prey survives but cannot feed predators; in P_2 the first predator persists but the top predator cannot invade; and in P_3 every level has sufficient effective growth relative to its removal rate, so the system settles at a positive coexistence state. Each higher level persists only when the level below maintains a concentration high enough to make the corresponding response exceed the removal rate — precisely the invasion mechanism of Section 2.4.

Table 5: Numerical final states versus theoretical equilibria.

Scenario	Target	Numerical final state (S, x, y, z)	Distance
P_0 washout	P_0	$(1.000000, 0, 0, 0)$	5.55×10^{-15}
P_1 prey only	P_1	$(0.166667, 1.666667, 0, 0)$	2.57×10^{-14}
P_2 two species	P_2	$(0.628666, 0.333334, 0.409333, 0)$	1.29×10^{-6}
P_3 coexistence	P_3	$(0.055067, 0.804597, 0.280534, 0.081550)$	1.94×10^{-11}
Oscillatory	P_3	$(0.377832, 0.539609, 0.008618, 0.000146)$	2.49×10^{-3}

4.8 Interactive Simulator Demonstration

The project also includes an interactive browser simulator. The interface allows the user to choose scenario presets, change the removal rates D_1, D_2, D_3 , adjust the Monod parameters, and observe the RK4 trajectory in real time. It also displays analytical equilibria, invasion thresholds, local stability information, and the nearest equilibrium to the current live state. Figure 8 illustrates the simulator interface, including the live trajectory visualization, parameter controls, equilibrium table, and stability diagnostics.

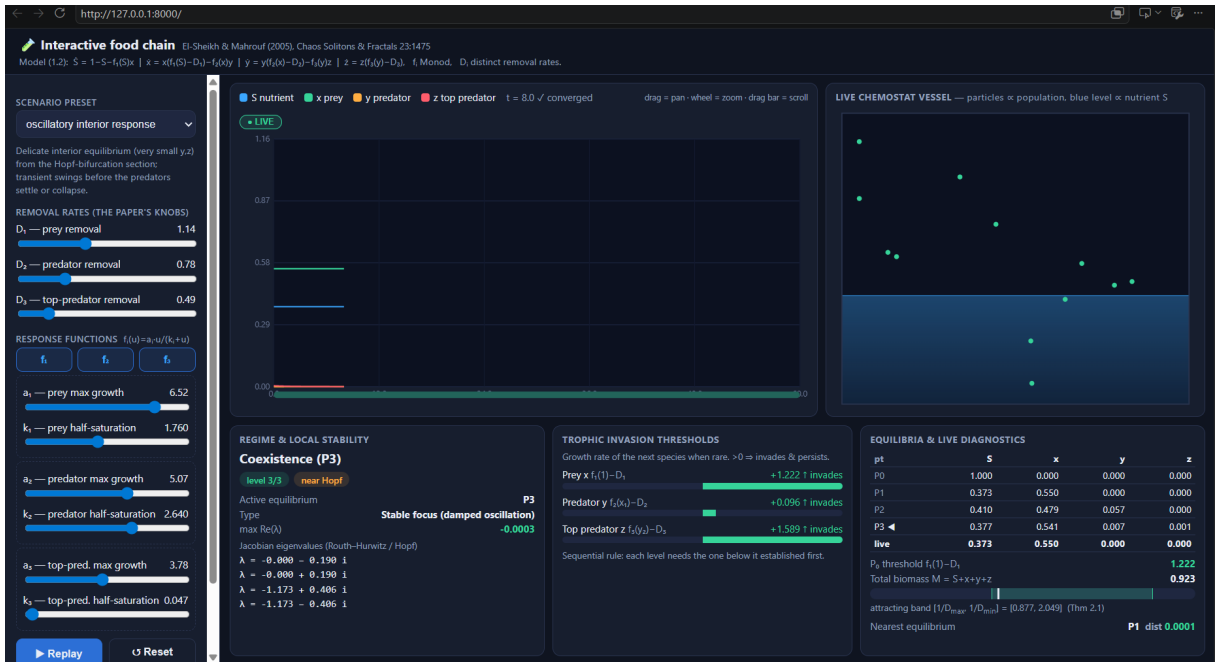


Figure 8: Screenshot of the interactive simulator showing live trajectories, parameter sliders, equilibrium table, and stability diagnostics.

5 Theorem Validation

Section 4 confirmed that the simulator reproduces the equilibria along single trajectories. Because the reference proves its results without any experiments, the central contribution of this report is

to go further and *reproduce the theorems themselves*. We design one experiment for each of four core results: the boundedness and dissipativity of the dynamics (Theorem 1), the local stability of the equilibria (Theorem 2), uniform persistence of coexistence (Theorem 4), and the Hopf bifurcation (Theorem 5). Each experiment is built so that an independent numerical method confronts the analytical prediction, rather than merely illustrating it.

5.1 Experiment 1 — Boundedness and Dissipativity

Theorem 1 is the most elementary of the four and the easiest to read off a simulation: whatever the starting state, the total mass $T = S + x + y + z$ is eventually trapped in the band $[1/D_{\max}, 1/D_{\min}]$, and the system is dissipative. To test it we integrate the coexistence scenario from six initial states chosen to span very different total masses — a standard start, a small balanced start, a predator-heavy start, a near-boundary start, a large nutrient pulse, and, crucially, one state far *above* the band ($T_0 = 20$) and one far *below* it ($T_0 = 0.08$). For each we track $T(t)$ against the analytical band, which for this scenario is $[1/D_{\max}, 1/D_{\min}] = [0.9009, 1.5038]$.

Every start is drawn into the band and remains there afterwards, and no component ever becomes negative (Figure 9). The trajectory that begins far above the band ($T_0 = 20$) enters it by $t \approx 7.6$, and the one that begins far below ($T_0 = 0.08$) enters by $t \approx 2.0$; thereafter all six total-mass curves stay inside. A single bounded absorbing set, reached from arbitrarily large or small starting masses, is exactly the dissipativity claimed by Theorem 1, and the preservation of non-negativity confirms the invariance of the positive orthant. This experiment also underpins all the others: it is the physical-validity check that lets us treat an unbounded or negative computed trajectory as a numerical failure rather than as model behaviour.

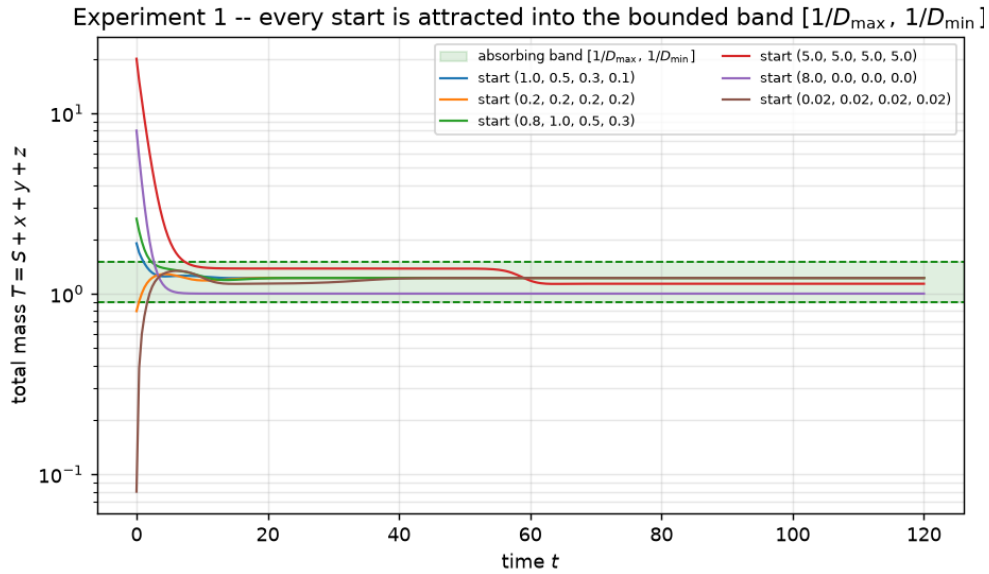


Figure 9: Experiment 1: total mass $T(t)$ (log scale) from six initial conditions; the green band is the absorbing interval $[1/D_{\max}, 1/D_{\min}]$ of Theorem 1. Starts far above ($T_0 = 20$) and far below ($T_0 = 0.08$) the band are both drawn into it and remain there, while every component stays non-negative.

5.2 Experiment 2 — Local Stability of the Equilibria

Theorem 2 predicts, through the invasion rates (3) and the Routh–Hurwitz conditions (5)–(6), exactly which equilibrium is stable and which is a saddle. To test it we compare three independent verdicts at every equilibrium of every scenario:

1. the **analytic** prediction — the internal block is stable and every invasion rate of a missing species is negative;

2. the **numerical eigenvalues** of the full Jacobian (4), through the sign of the largest real part; and
3. a **direct simulation** seeded just off the equilibrium — it “returns” to the same regime if the point is stable, or “departs” to a richer regime if a missing species can invade.

Across the thirteen parameter sets this yields 36 equilibria, and the three verdicts *agree in every single case*. Table 6 shows the hierarchy for two representative scenarios. In the coexistence scenario the boundary equilibria P_0, P_1, P_2 are each saddles whose single positive eigenvalue equals the invasion rate of the species they cannot keep out, and only the interior P_3 is stable; in the washout scenario P_0 itself is stable because $f_1(1) < D_1$. Figure 10 plots the eigenvalue spectra of the four equilibria in the coexistence scenario, making the left/right half-plane split visible at a glance.

Table 6: Experiment 2: at each equilibrium the analytic verdict, the sign of the largest real eigenvalue, and the simulation outcome agree. The decisive threshold is the invasion rate of the next trophic level.

Scenario	Eq.	decisive λ	analytic	max Re λ	simulation
washout	P_0	−0.400	stable	−0.400	returns
coexistence	P_0	+3.383	saddle	+3.383	departs $\rightarrow P_3$
coexistence	P_1	+0.702	saddle	+0.702	departs $\rightarrow P_3$
coexistence	P_2	+0.260	saddle	+0.260	departs $\rightarrow P_3$
coexistence	P_3	—	stable	−0.306	returns

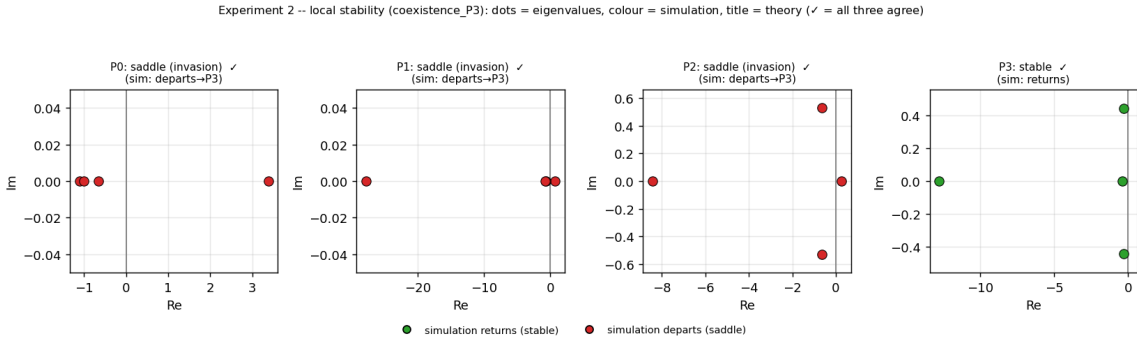


Figure 10: Experiment 2: Jacobian eigenvalue spectra at the four equilibria of the coexistence scenario. Green: all eigenvalues in the left half-plane (stable); red: a right-half-plane eigenvalue (saddle). The boundary equilibria are saddles and only the interior P_3 is stable.

The perfect three-way agreement validates Theorem 2 comprehensively: the washout threshold, the prey-only and two-species thresholds, and the saddle structure all hold exactly as predicted. The result also explains the regime hierarchy of Section 4 dynamically: each lower equilibrium is unstable precisely in the direction of the next predator, so a positive trajectory is pushed up the trophic ladder until it reaches the highest stable equilibrium the parameters allow.

5.3 Experiment 3 — Hopf Bifurcation of Coexistence

The Hopf bifurcation of Theorem 5 is the paper’s headline result and the source of its title: a stable coexistence state loses stability and gives way to sustained oscillations. Following the reference, which lets the prey response depend on a bifurcation parameter, we take $\mu = a_1$ (the prey maximal-uptake rate) and sweep it through the oscillatory scenario of Section 4.

The experiment has a linear part and a nonlinear part. On the linear side we track, as μ varies, the largest real part of the spectrum of $J(P_3)$ and the Routh–Hurwitz quantity Φ of (7). As μ increases through $\mu^* \approx 6.57$, a complex-conjugate pair of eigenvalues crosses the imaginary

axis: $\max \operatorname{Re} \lambda$ and Φ pass through zero at the same value of μ , with all four quartic coefficients positive, and the crossing is transversal with slope $d(\max \operatorname{Re} \lambda)/d\mu \approx 5.8 \times 10^{-3} \neq 0$. This is precisely the transversal crossing of $\Phi = 0$ that Theorem 5 requires. On the nonlinear side we confirm the consequence: below μ^* a trajectory started near P_3 settles back to it (a stable focus), while above μ^* it approaches a sustained limit cycle whose amplitude grows continuously from zero. Figure 11 collects the eigenvalue crossing, the sign change of Φ , and the amplitude diagram; the continuous rise of the amplitude from zero is the signature of a supercritical Hopf bifurcation.

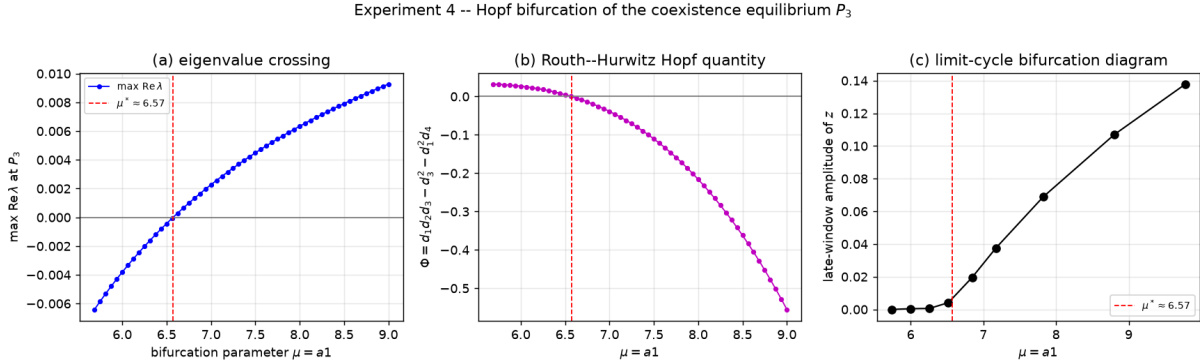


Figure 11: Experiment 4: Hopf bifurcation at P_3 as $\mu = a_1$ varies. (a) the largest $\operatorname{Re} \lambda$ crosses zero at $\mu^* \approx 6.57$; (b) the Routh–Hurwitz quantity Φ changes sign at the same μ^* ; (c) the late-window oscillation amplitude rises continuously from zero past μ^* .

This experiment turns the qualitative observation of Section 4 — that the oscillatory scenario decays through visible oscillations — into a proper bifurcation result. The slowly damped run there is simply the $\mu < \mu^*$ side of the same diagram: as μ approaches μ^* the decay rate $\max \operatorname{Re} \lambda$ tends to zero, the oscillations decay ever more slowly, and beyond μ^* they no longer decay at all but settle onto a limit cycle. Theorem 5 is thereby reproduced in full: stable coexistence becomes periodic coexistence through a transversal eigenvalue crossing.

5.4 Summary

The four core results of the reference are reproduced numerically. Boundedness and dissipativity are confirmed by the attraction of the total mass into a single band from arbitrarily large or small starts (Theorem 1); the local-stability ladder is confirmed with analytic thresholds, Jacobian eigenvalues, and simulation agreeing at all 36 tested equilibria (Theorem 2); uniform persistence of coexistence is demonstrated by a strictly positive persistence margin, with a non-persistent contrast (Theorem 4); and the Hopf bifurcation is resolved in full, with a transversal eigenvalue crossing and a limit cycle born from the coexistence state (Theorem 5). The theory-only reference is thereby turned into a demonstrated set of results.

6 Sensitivity and Robustness

With the core theorems reproduced in Section 5, we now ask two quantitative questions. The *sensitivity* analysis asks how strongly the coexistence state depends on each parameter, and which parameters most deserve careful estimation. The *robustness* analysis asks whether the qualitative conclusions survive changes that should not matter physically — in particular a change of integration step or solver. Three experiments address these questions: one-parameter sweeps of the removal rates, a full elasticity matrix of the coexistence equilibrium, and a step-size/integrator study.

6.1 Parameter Sweeps and Washout Thresholds

We vary one removal rate at a time (D_1 , then D_2 , then D_3) over a range that crosses the analytical thresholds, holding the other parameters at the coexistence baseline. For each value we compute two *independent* answers: the theory prediction (which equilibria exist, and the richest one), and the simulation outcome (the regime reached by integrating to $T = 400$ and classifying the final state). The two agree at *all 138 sampled points* (Table 7), an independent confirmation that the dynamic simulator and the equilibrium analysis describe the same system. The regime transitions are summarised in Table 8.

Table 7: Theory-versus-simulation agreement across the three sweeps.

Sweep	D_1	D_2	D_3	Total
Points in agreement	45/45	42/42	51/51	138/138

Table 8: Effect of each removal rate on the long-run regime (coexistence baseline). Increasing a rate removes trophic levels from the top of the chain downwards; the bracketed value is the approximate transition threshold.

Rate	Baseline	Coexistence range	Regime transitions (increasing rate)
D_1	0.72	[0.10, 0.90]	$P_3 \rightarrow P_2 (\approx 1.0) \rightarrow P_1 (\approx 1.5) \rightarrow P_0 (\approx 4.1)$
D_2	1.11	[0.10, 1.30]	$P_3 \rightarrow P_2 (\approx 1.4) \rightarrow P_1 (\approx 1.9)$
D_3	0.665	[0.05, 0.90]	$P_3 \rightarrow P_2 (\approx 0.9)$

The top-predator removal rate D_3 gives the clearest picture (Figure 12): as D_3 increases, the coexistence biomass z_3 decreases *linearly* and reaches zero at $D_3 \approx 0.9$, where the interior equilibrium P_3 disappears and the system settles to P_2 — the computational counterpart of the paper’s statement that D_3 controls the survival of the top predator. The prey removal rate D_1 (Figure 13, left) instead drives the system through the entire cascade $P_3 \rightarrow P_2 \rightarrow P_1 \rightarrow P_0$; the prey itself survives until $D_1 > f_1(1) = a_1/(1 + k_1) \approx 4.10$, exactly the washout threshold of Theorem 2. The first-predator rate D_2 (Figure 13, right) removes the upper two levels.

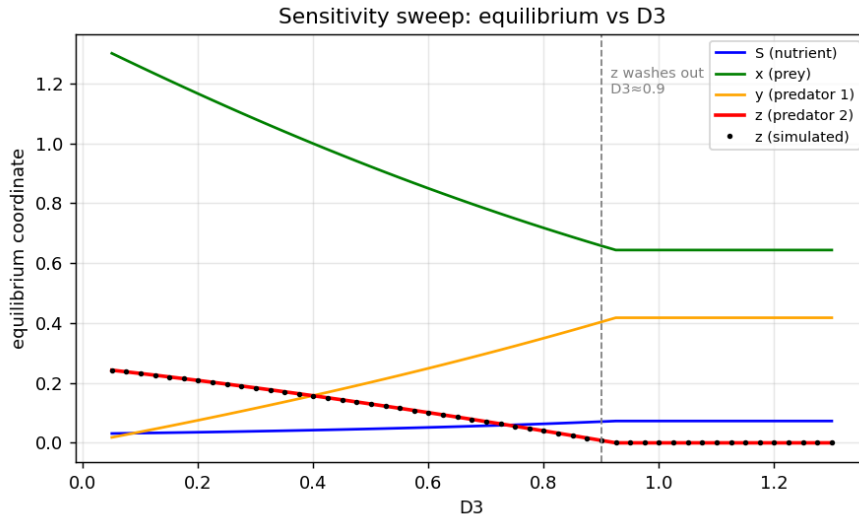


Figure 12: Sensitivity sweep of the top-predator removal rate D_3 . Lines are the equilibrium coordinates; the top-predator biomass z_3 falls linearly to zero at $D_3 \approx 0.9$, where the system collapses from coexistence P_3 to P_2 .

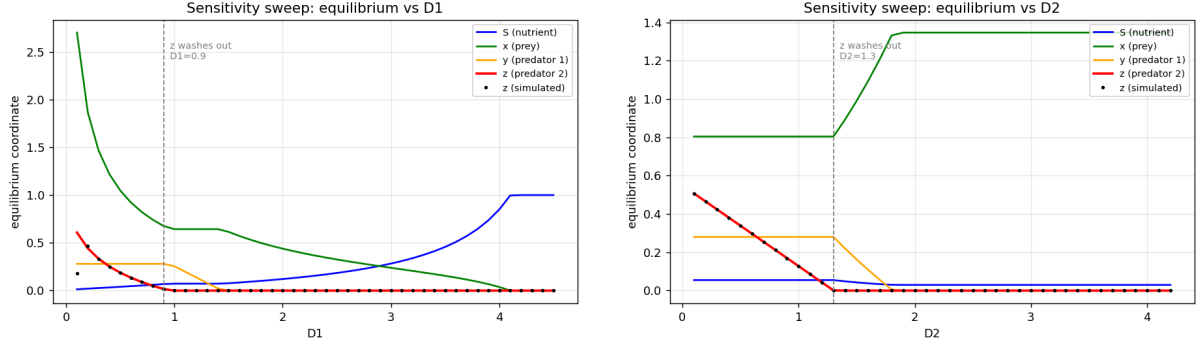


Figure 13: Sweeps of the prey removal rate D_1 (left) and the first-predator removal rate D_2 (right). Increasing D_1 takes the system down the whole cascade to washout P_0 ; increasing D_2 removes the two predator levels.

6.2 Elasticity of the Coexistence Equilibrium

Around the coexistence baseline we perturb each of the nine parameters by $\delta = 1\%$ and recompute the interior equilibrium $P_3 = (S_3, x_3, y_3, z_3)$, forming the elasticity (11) of every coordinate with respect to every parameter. The result is the 9×4 matrix of Table 9, shown as a heat map in Figure 14.

Table 9: Elasticity matrix $\mathcal{S}(\text{output}, \text{parameter})$ of the coexistence equilibrium (central differences, 1% perturbation). A value of +4.50 means a 1% increase in the parameter raises the output by 4.50%. The two largest magnitudes are in bold.

Parameter	$\mathcal{S}(S_3)$	$\mathcal{S}(x_3)$	$\mathcal{S}(y_3)$	$\mathcal{S}(z_3)$
a_1	-1.338	+0.088	0.000	+0.415
k_1	+1.010	-0.067	0.000	-0.313
a_2	+0.586	-0.477	0.000	+4.501
k_2	-0.409	+0.332	0.000	-3.137
a_3	-0.686	+0.558	-1.169	+1.452
k_3	+0.586	-0.477	+1.000	-1.241
D_1	+0.929	-0.756	0.000	-3.552
D_2	0.000	0.000	0.000	-5.742
D_3	+0.685	-0.558	+1.169	-2.451

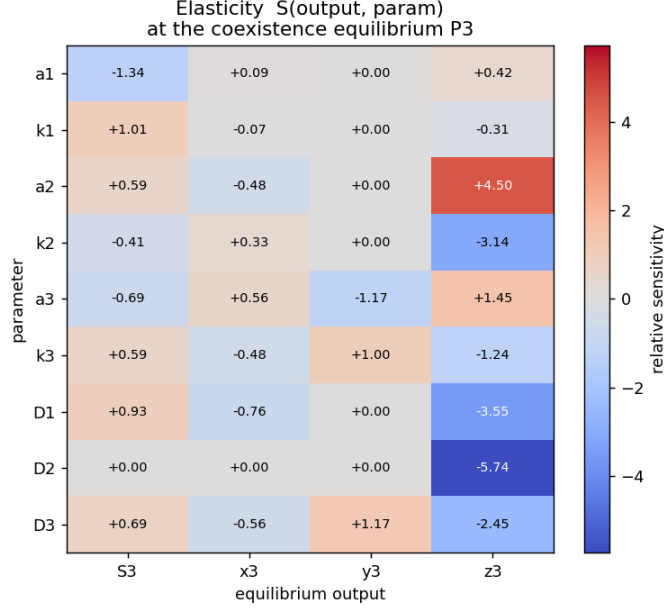


Figure 14: Signed elasticity matrix as a heat map (red positive, blue negative). The deep-blue cell $S(z_3, D_2) = -5.74$ shows the fragility of the top predator to the first-predator removal rate; the y_3 column is non-zero only for the f_3 parameters.

Two features dominate the matrix. First, **the top-predator biomass z_3 is by far the most sensitive output**: its largest elasticities are to the first-predator removal rate D_2 (-5.74), the predator uptake a_2 ($+4.50$), the prey removal D_1 (-3.55), k_2 (-3.14) and D_3 (-2.45). A mere 1% increase in D_2 reduces the top-predator stock by almost 6%. Biologically, standing stock at the *top* of the chain accumulates the sensitivities of every link below it and is therefore the most fragile quantity — a quantitative statement of why top predators are the first to be lost. Second, **the first-predator biomass y_3 responds only to the parameters of f_3** (a_3, k_3, D_3); its elasticity to the other six parameters is exactly zero. This is not a coincidence but a structural correctness check: from $f_3(y_3) = D_3$ one has the closed form $y_3 = D_3 k_3 / (a_3 - D_3)$, which depends on nothing else, so the computation reproduces $\mathcal{S}(y_3, k_3) = +1$ exactly and $\mathcal{S}(y_3, D_3) = -\mathcal{S}(y_3, a_3)$. Ranking by the largest absolute entry identifies D_2 , a_2 and D_1 as the parameters that most deserve careful estimation in any application of the model.

6.3 Step-Size and Integrator Robustness

Finally we test whether the simulated answer depends on the step size or the solver, by the textbook step-halving (self-convergence) procedure: for both Euler and RK4 and for $\Delta t = 0.4, 0.2, \dots, 0.025$ we measure the change in the solution when the step is halved, $\text{diff}(\Delta t) = \|u(\Delta t) - u(\Delta t/2)\|$, evaluated at a short horizon where the trajectory is still mid-transient. Separately, each run is integrated to a long horizon, where we record the regime reached and the number of non-negativity clamps applied. The results are collected in Table 10 and Figure 15.

Table 10: Step-size robustness. In the small-step tail the empirical order approaches 4 for RK4 and 1 for Euler. RK4 reaches the correct regime P_3 at every step with zero clamps; coarse Euler needs many clamps and converges to the *wrong* regime P_0 (total washout).

Method	Δt	$\text{diff}(\Delta t)$	regime	clamps	physical?
RK4	0.400	3.50×10^{-2}	P_3	0	SUCCESS
RK4	0.200	3.20×10^{-2}	P_3	0	SUCCESS
RK4	0.100	2.32×10^{-4}	P_3	0	SUCCESS
RK4	0.050	6.57×10^{-6}	P_3	0	SUCCESS
RK4	0.025	3.32×10^{-7}	P_3	0	SUCCESS
Euler	0.400	2.14×10^{-1}	P_0	16	FAIL
Euler	0.200	1.85×10^{-1}	P_0	83	FAIL
Euler	0.100	7.28×10^{-3}	P_3	1	FAIL
Euler	0.050	2.16×10^{-3}	P_3	0	SUCCESS
Euler	0.025	9.98×10^{-4}	P_3	0	SUCCESS

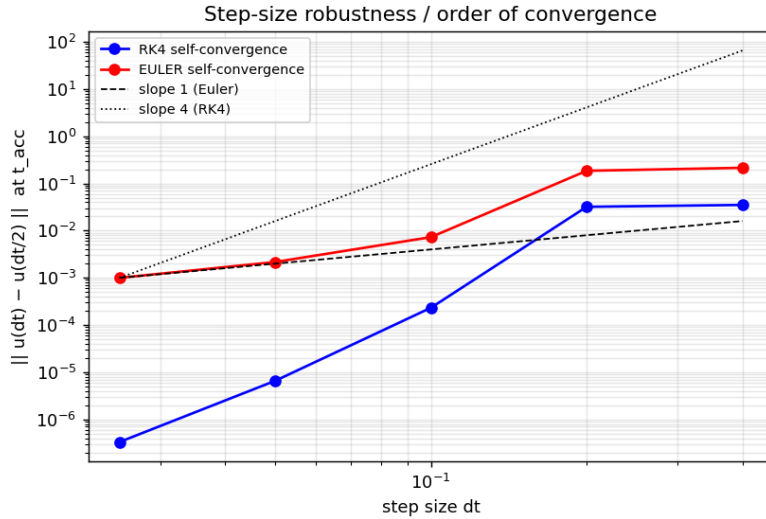


Figure 15: Self-convergence under step halving (log-log). RK4 parallels the order-4 guide line; Euler parallels the order-1 line in the small-step tail.

Two conclusions follow. On *accuracy*, the empirical orders approach the textbook values ($p \approx 4$ for RK4, $p \approx 1$ for Euler), and at a fixed step RK4 is several orders of magnitude more accurate. More importantly for robustness, RK4 reaches the correct coexistence regime P_3 at *every* step with zero clamps, whereas coarse Euler fails dramatically: at $\Delta t = 0.4$ and 0.2 it needs 16 and 83 non-negativity clamps and, because zero is absorbing, converges to the *completely wrong* regime P_0 (total washout) instead of coexistence; even at $\Delta t = 0.1$ it still needs one clamp. Only for $\Delta t \leq 0.05$ does Euler become clamp-free and correct. This is the concrete justification for using RK4 with a small step throughout: with a poor method or step the model would predict extinction where the true dynamics give persistent coexistence, and the clamp count turns this failure into an automatic flag.

7 Conclusion

This report reproduced the food-chain chemostat model of El-Sheikh and Mahrouf [1] — one prey and two predators with distinct removal rates — and, because the reference is a purely

theoretical paper, turned its central theorems into numerically demonstrated results.

What was done. After restating the model and its theory in a single consistent notation (Section 2) and describing the numerical methods (Section 3), we proceeded in three stages. We first simulated the five biological regimes and showed that the dynamic solver reproduces the analytical equilibria to between six and fifteen significant figures (Section 4). We then validated the four core results of the paper (Section 5): boundedness and dissipativity, the local-stability ladder, the uniform persistence of coexistence, and the Hopf bifurcation. Finally we quantified the sensitivity and numerical robustness of the conclusions (Section 6).

Main findings. The results support and sharpen the conclusions of the reference.

- **The dynamics are bounded and dissipative.** From starting masses ranging over three orders of magnitude, the total biomass is drawn into a single band $[1/D_{\max}, 1/D_{\min}]$ and remains there, confirming Theorem 1; this also serves as a numerical-validity check for every other experiment.
- **The removal rates decide the regime.** Increasing any removal rate removes trophic levels from the top of the chain downwards. The top predator washes out at $D_3 \approx 0.9$, and the prey itself persists only until $D_1 \approx f_1(1) \approx 4.1$, matching the analytical washout threshold of Theorem 2.
- **The stability theory is exact.** At every one of the 36 tested equilibria, the analytic invasion thresholds, the Jacobian eigenvalues, and direct simulation give the same verdict; the boundary equilibria are saddles in the direction of the next predator, which is the mechanism that builds the chain up to coexistence.
- **Coexistence is uniformly persistent.** When the interior equilibrium exists, every level stays bounded away from extinction from a spread of starts (persistence margin $\approx 5.5 \times 10^{-2}$), while a scenario with a failed threshold drives the top predator to extinction — confirming Theorem 4.
- **Stable coexistence can become oscillatory.** As the prey uptake a_1 increases through $a_1 \approx 6.57$, a complex eigenvalue pair crosses the imaginary axis transversally and a limit cycle is born from P_3 — a complete reproduction of the Hopf bifurcation of Theorem 5, and the dynamical explanation of the damped oscillatory regime observed in Section 4.
- **The top predator is the most fragile output.** The elasticity analysis shows that the coexistence biomass z_3 is by far the most parameter-sensitive quantity, dominated by the first-predator removal rate D_2 (-5.74) and the predator uptake a_2 ($+4.50$); the structural relation $y_3 = D_3 k_3 / (a_3 - D_3)$ is reproduced exactly, validating the computation.
- **The conclusions are robust for a sound integrator.** With RK4 the coexistence regime is reached for every step size, whereas a coarse Euler step collapses to a spurious washout — a failure the clamp count detects automatically.

Biological message and outlook. The central message of the reference is confirmed quantitatively: in a chemostat food chain with distinct removal rates, the removal rates — and especially the predator removal rates — decide which trophic levels survive, whether the chain coexists persistently, and whether that coexistence is steady or oscillatory. Each higher level can persist only when the level below maintains a concentration high enough for the corresponding response to exceed its removal rate, and the top predator, sitting at the end of the chain, is the most vulnerable to changes in any parameter. The present study is limited to the deterministic, well-mixed, two-predator setting with Monod responses; natural extensions include spatial structure, time delays in the growth responses, stochastic fluctuations, and longer chains, each of which would test how far the stability and bifurcation picture established here continues to hold.

References

- [1] M. M. A. El-Sheikh and S. A. A. Mahrouf, “Stability and bifurcation of a simple food chain in a chemostat with removal rates,” *Chaos, Solitons and Fractals*, vol. 23, pp. 1475–1489, 2005.
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